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HTML QUICK-STYLER

AI-Powered HTML Page Generator using IBM Granite 3.3 2B

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ABSTRACT

Web development often requires significant time and technical expertise to design professional and responsive websites. Many designers and developers spend considerable effort creating layouts, selecting color schemes, and writing structured HTML and CSS code. To simplify this process, the HTML Quick-Styler system provides an AI-powered platform capable of automatically generating modern web pages.

The system uses the IBM Granite 3.3 2B Instruct model to analyze user descriptions such as website title, content requirements, and preferred color styles. Based on this input, the platform generates a complete HTML structure, CSS styling, responsive layout, and design elements automatically.

The application is built using Python, Gradio, HTML5, CSS3, and JavaScript, and it is deployed using Google Colab for easy accessibility. The interface allows users to generate pages, preview designs, explore color palettes, and view template structures through a simple interactive interface.

INTRODUCTION

Web development is an essential part of modern digital communication. Businesses, designers, and organizations require websites to present information and services to users. However, creating a professional website requires knowledge of HTML, CSS, layout design, and responsive development techniques.

The HTML Quick-Styler project is designed to simplify the web development process by providing an AI-powered tool that automatically generates structured and responsive web pages.
The system analyzes user inputs and generates:

- Complete HTML page structure
- Professional CSS styling
- Responsive layout for multiple devices
- Animations and modern UI components

By automating these tasks, the system allows users to build websites quickly without requiring deep technical expertise.

OBJECTIVES

The main objectives of this project are:

- To develop an AI-powered system for generating HTML web pages.
- To reduce the time required for website development.
- To automatically create responsive layouts and modern designs.
- To provide multiple color palette options for design customization.
- To offer an easy-to-use interface through the Gradio framework.

Scope of the Project

The system can be used by:

- Web designers
- Developers
- Students learning web development
- Startups creating landing pages
- Businesses needing quick website prototypes

The platform supports generation of several website components such as hero sections, portfolio galleries, contact forms, and service listings.

SYSTEM ANALYSIS

Existing System

In the existing approach, web pages are developed manually by writing HTML and CSS code. Designers must also create layouts, choose color schemes, and ensure compatibility across devices.

Limitations of Existing System

- Requires strong coding knowledge
- Time-consuming development process
- Manual design selection
- Difficulty maintaining consistent styling
- Limited automation

Proposed System

The proposed system introduces an AI-based HTML generator that automatically creates web pages from user descriptions.

Users provide:

- Website title
- Page description
- Color style preference

The system then generates:

- HTML structure
- CSS styling
- Responsive design
- Section layouts

Advantages of Proposed System

- Faster web page generation
- Reduced manual coding
- Modern design templates
- Responsive layouts
- Easy user interface

SYSTEM DESIGN

System Architecture

The architecture of HTML Quick-Styler consists of several modules working together.

Main components include:

1. User Interface (Gradio)
2. Python Backend
3. AI Model (IBM Granite 3.3 2B)
4. HTML Generator
5. CSS Styling Engine
6. Web Page Preview System

Workflow:

User Input → AI Model → HTML & CSS Generation → Preview Output

Core Technologies

The project uses the following technologies:

- Python for application logic
- IBM Granite 3.3 2B Instruct Model for AI-based content generation
- Gradio for user interface development
- HTML5 and CSS3 for webpage creation
- JavaScript for interactive features
- Google Colab for deployment

Color Palette Generator

The system includes a module called ColorPaletteGenerator which provides predefined color themes.

Available themes include:

- Modern
- Vibrant
- Ocean
- Forest
- Sunset

IMPLEMENTATION

Phase 1 – Environment Setup

Tasks performed:

- Creating a Google Colab notebook
- Installing required libraries
- Enabling GPU runtime
- Loading the IBM Granite AI model

Phase 2 – Core Application Development

Key components developed:

Color Palette Generator

Generates predefined color themes.

HTML Page Builder

Detects webpage sections and creates structured HTML code.

CSS Generator

Creates styling rules, animations, and responsive layouts.

Section Templates

Includes templates for:

- Hero section
- Features section
- Portfolio gallery
- Pricing table
- Testimonials
- Contact form
- Footer

Phase 3 – Gradio Interface

The system interface includes five tabs.

1. Generate Page
2. Live Preview
3. Color Palettes

TESTING AND RESULTS

Functional testing was conducted to ensure the system generates correct web pages.

Test Case

Input:

Title: Modern Portfolio Website

Description: Hero section, features list, about section, portfolio gallery, testimonials, contact form, and footer.

Color Style: Modern

Expected Output:

- Fully styled responsive web page
- Structured HTML layout
- CSS styling and animations

Result

The system successfully generates professional web pages based on user descriptions.

Generated output includes:

- Responsive HTML layout
- Modern CSS styling
- Organized section structure
- Ready-to-deploy webpage code

FUTURE ENHANCEMENTS

Possible improvements include:

- Custom color picker feature
- Advanced JavaScript component generation
- Integration with frontend frameworks
- Additional design templates
- Export options for different web platforms

CONCLUSION

The HTML Quick-Styler project demonstrates the effectiveness of artificial intelligence in simplifying web development.

By integrating the IBM Granite AI model with a Python backend and Gradio interface, the system provides an efficient solution for generating professional web pages automatically.

The platform reduces development time and allows users to create modern websites without extensive coding knowledge.